On the Stack Sortable and Pushpopable Permutations *

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Abstract

This work proposes an $O(\log n)$ time parallel algorithm for testing whether a permutation is stack sortable or not by $O(n)$ processors under the PRAM EREW computational model. Additionally, the number of permutations which are both stack sortable and pushpopable is also discussed, and an equality is finally obtained, which combines $n!$, 2^n, C_n, and U_n together where C_n is the Catalan number and U_n is the number of permutations which are both stack unsortable and unpushpopable.

Keyword: Permutations, stack sortable permutations, inverse permutations, inversion tables, involution permutations, Catalan numbers.

1 Introduction

Let $\Pi = p_1p_2\cdots p_n$ be a permutation on $\{1, 2, \cdots, n\}$. Permutation Π is called a *stack sortable permutation* if there is a way to obtain the identity permutation $12\cdots n$ from Π by using a stack and the push/pop operations. For example, 123, 132, 213, 312, and 321 are stack sortable permutations when $n = 3$. However, permutation 231 is not a stack sortable permutation. A permutation is a stack sortable permutation if and only if it avoids the (scattered) pattern 2-3-1 [10]. A 2-3-1 (resp. 3-1-2) pattern in a permutation $\Pi = p_1p_2\cdots p_n$ means that there exist p_i, p_j, and p_k with $i < j < k$ and $p_j > p_i > p_k$ (resp. $p_i > p_k > p_j$). Oppositely, Π is called a *pushpopable permutation* if there is a way to obtain Π from the identity permutation $12\cdots n$ by using a stack and the push/pop operations [4]. For example, 123, 132, 213, 231, and 321 are pushpopable permutations when $n = 3$. However, permutation 312 is not a pushpopable permutation. A permutation is a pushpopable permutation if and only if it avoids the (scattered) pattern 3-1-2 [4]. In [1], Albert and Atkinson gave a linear time algorithm for testing whether a permutation is stack sortable or not. This work proposes an $O(\log n)$ time parallel algorithm for testing a stack sortable permutation by using $O(n)$ processors under the PRAM EREW computational model. Besides, we are also concerned with the number of permutations which are both stack sortable and pushpopable. Finally, an equality which combines $n!$, 2^n, C_n, and U_n together is obtained, where C_n is the Catalan number and U_n is the number of permutations which are both stack unsortable and unpushpopable.

The remaining part of this study is organized as follows. Section 2 introduces some properties on permutations. After that, an $O(\log n)$ time parallel algorithm is proposed for determining whether a permutation is stack sortable or not. Section introduces two new integer sequences. One of them is equal to $n! - 2C_n + 2^{n-1}$ and the other is equal to $n! - 2C_n - I_n + 2^n$, where C_n is the Catalan number, U_n is the number of stack unsortable and unpushpopable permutations, and I_n is the number of involution permutations.

2 A Parallel Algorithm for Determining a Sortable Permutation

An *inverse* of permutation Π is a permutation in which each number of Π and the number of

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the place which it occupies are exchanged [8]. For example, $\Pi_1 = 231$ and $\Pi_2 = 312$ are inverse permutations of each other since the positions of 1, 2, and 3 in Π_1 are 3, 1, and 2, respectively, namely Π_2 and the positions of 1, 2, and 3 in Π_2 are likewise Π_1. Let Π^{-1} denote the inverse of permutation Π. Hence, if $\Pi = 231$, then $\Pi^{-1} = 312$ and vice versa. A pair of elements (p_i, p_j) is called an *inversion* in a permutation Π if $i < j$ and $p_i > p_j$. Corresponding to a permutation $p_1p_2 \cdots p_n$ there is a sequence of integers $t_1t_2 \cdots t_n$ such that $t_1 = 0$ and for $2 \leq i \leq n$, t_i is the number of elements in $\{p_1, p_2, \cdots, p_{i-1}\}$ which are greater than p_i. The sequence $t_1t_2 \cdots t_n$ is called the *inversion table* for the permutation $p_1p_2 \cdots p_n$. A permutation determines a unique inversion table. Furthermore, given an inversion table, its associated permutation can be reconstructed in $O(n \log n)$ time [7]. If a permutation is a *tree permutation*, i.e., $t_{i+1} - t_i < 2$ for all $i = 1, 2, \cdots, n-1$, then there is a linear time algorithm for converting it to its inversion table and vice versa [6]. If permutation $\Pi = p_1p_2 \cdots p_n$ and its corresponding inversion table is $t_1t_2 \cdots t_n$, then $C = c_1c_2 \cdots c_n$ is called the *permusion* of Π where $c_i = p_i + t_i$ for $i = 1, 2, \cdots, n$. A permusion is *nondecreasing* if $c_i \leq c_{i+1}$ for $i = 1, 2, \cdots, n-1$. For example, permutation 231 contains two inversions $(2, 1)$ and $(3, 1)$ and yields the inversion table 002. The permusion of 231 is 233 and it is a nondecreasing permusion.

In the following, a *231-avoid* (respectively, *312-avoid*) permutation is adopted to represent a permutation without the (scattered) pattern 2-3-1 (respectively, 3-1-2).

Lemma 1 *[10] A permutation is a stack sortable permutation if and only if it is 231-avoid.*

Lemma 2 *[4] A permutation is a pushpopable permutation if and only if it is 312-avoid.*

Lemma 3 *Permutation* Π *is a stack sortable permutation if and only if* Π^{-1} *is a pushpopable permutation.*

Proof. Since 231 and 312 are the inverse permutations of each other, Π is 231-avoid if and only if Π^{-1} is 312-avoid. According to Lemmas 1 and 2, this lemma holds. Q. E. D.

Lemma 4 *Permutation* Π *is a pushpopable permutation if and only if its permusion is nondecreasing.*

Proof. The permusion of 3-1-2 is 3-2-3, which is not nondecreasing. Hence, if the permusion of a permutation Π is nondecreasing, then Π is 312-avoid and Π is a pushpopable permutation consequently by Lemma 2.

On the other hand, suppose on the contrary that there is a pushpopable permutation Π with not nondecreasing permusion $c_1c_2 \cdots c_n$, say $c_k > c_{k+1}$ for some $k \in \{1, 2, \cdots, n-1\}$. Note that $c_i = p_i + t_i$ for all $i \in \{1, 2, \cdots, n\}$. If $p_k < p_{k+1}$ and $p_{k+1} - p_k = g$ for some positive integer g, then this implies $t_k - t_{k+1} \geq g + 1$ for $c_k = p_k + t_k > p_{k+1} + t_{k+1} = c_{k+1}$. This means that there exist at least $g + 1$ numbers greater than p_k and smaller than p_{k+1} in $\{p_1, p_2, \cdots, p_{k-1}\}$. It is impossible since $p_{k+1} - p_k = g$. Hence, $p_k > p_{k+1}$. Then $t_k < t_{k+1}$ by the definition of inversion table $t_1t_2 \cdots t_n$. Suppose $t_{k+1} - t_k = g$ for some positive integer g, then $p_k - p_{k+1} = (c_k - t_k) - (c_{k+1} - t_{k+1}) \geq g + 1$. Therefore, if there exists some positive integer $j \in \{k+2, k+3, \cdots, n\}$ such that $p_j = p_{k+1} + x$ for some $1 \leq x \leq g$, and $p_k - p_{k+1} - p_j$ forms a 3-1-2 pattern, a contradiction; otherwise, $t_{k+1} - t_k = p_k - p_{k+1} \geq g$, a contradiction, too. Q. E. D.

We are now in a position to describe our parallel algorithm for determining whether a permutation is stack sortable or not.

Algorithm A Check if Π is stack sortable permutation or not.

Input: A permutation $\Pi = p_1p_2 \cdots p_n$.

Output: YES, if Π is a stack sortable permutation; NO, otherwise.

Method:

Step 1. Find $\Pi^{-1} = p'_1p'_2 \cdots p'_n$.

Step 2. Get n points $(i, n+1-p'_i)$ in the plane for $i = 1, 2, \cdots, n$.

Step 3. Apply the method in [2] to compute t_i, the number of points which are located in the lower left side of each point $(i, n+1-p'_i)$ in the plane for $i = 1, 2, \cdots, n$.

Step 4. For $i = 1, 2, \cdots, n$, let $c_i = p'_i + t_i$.

Step 5. If there exists some $i \in \{1, 2, \cdots, n-1\}$ such that $c_i > c_{i+1}$, then output **NO**. Otherwise, output **YES**.

End of Algorithm A

An example is used to illustrate *Algorithm A*. For example, $\Pi = 4312$. In Step 1, $\Pi^{-1} = 3421$ is obtained. In Step 2, four points (1, 2), (2, 1), (3, 3), and (4, 4) are added in the plane (See Figure 1). A point a is *dominated* by another point b if a is located in the lower left side of b in the plane, i.e., both x-coordinate and y-coordinate of point a are less than those of point b. In Step 3, the numbers of points dominated by each point are 0, 0, 2, and 3 can be determined, respectively, of which the inversion table of 3421 is composed. The permusion of 3421 is 3444 which is calculated in Step 4. Obviously, 3444 is in nondecreasing order and the answer is **YES** in Step 5. Conversely, if $\Pi = 3421$, then $\Pi^{-1} = 4312$. The permusion of 4312 is 4434 which is not a nondecreasing permusion. Hence, the answer of Algorithm A will be **No**. Note that, it is easy to see that delete Step 1 and replace Π^{-1} by Π in the following steps will change Algorithm A to be an algorithm which can determine whether a permutation is pushpopable or not.

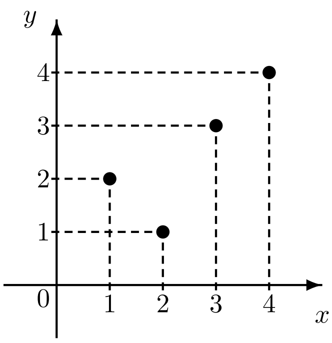

Figure 1: Point transformation and dominating points.

Theorem 5 *Algorithm A determines in parallel whether a permutation is stack sortable or not in $O(\log n)$ time by using $O(n)$ processors under the PRAM EREW computational model.*

Proof. The correctness of Algorithm A can be validated by Lemmas 3 and 4. The time-complexity of Algorithm A is analyzed as follows. In Step 1, finding the inverse permutation of Π can be done in $O(1)$ time by using $O(n)$ processors. Step 2 also takes $O(1)$ time for creating n points in the plane by using $O(n)$ processors. In [2], Atallah et. al. gave an $O(\log n)$ time parallel algorithm for finding the number of points dominated by all points in the plane in an EREW PRAM computational model with $O(n)$ processors. Steps 4 and 5 can also be done in $O(1)$ time by using $O(n)$ processors. Therefore, the parallel time-complexity of Algorithm A is $O(\log n)$ by using $O(n)$ processors in an EREW PRAM computational model. Q. E. D.

3 The Number of Stack Sortable and Pushpopable Permutations

A permutation Π is called an *involution permutation* if $\Pi = \Pi^{-1}$ (i.e, a permutation that is its own inverse permutation). For example, the unique involution permutation on $n = 1$ is 1, the two involution permutations on $n = 2$ are 12 and 21, and the four involution permutations on $n = 3$ are 123, 132, 213, and 321. In general, the number of involution permutations on n numbers is given by the recurrence

$$I_n = I_{n-1} + (n-1)I_{n-2}$$

or alternatively by

$$I_n = n! \cdot \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \frac{1}{2^k k!(n-2k)!}$$

[5, 9].

A set of subpermutations $\Pi_1, \Pi_2, \cdots, \Pi_k$ is called a *proper separation* of permutation $\Pi = p_1 p_2 \cdots p_n$ if $\Pi = \Pi_1 \circ \Pi_2 \circ \cdots \circ \Pi_k$ where $\circ$ is a concatenating operation. A subpermutation is called a *decreasing subpermutation* if the numbers appeared in it are in decreasing order. A proper separation is called an *increasing separation* if each subpermutation of it is a decreasing subpermutation and, for any $i < j$, every number in Π_j is greater than any number in Π_i. For example, $\Pi = 132654$ can be separated into three subpermutations: $\Pi_1 = 1, \Pi_2 = 32$, and $\Pi_3 = 654$, each of them is a decreasing subpermutation. Clearly, $\Pi = \Pi_1 \circ \Pi_2 \circ \Pi_3$ and the set of subpermutations Π_1, Π_2, and Π_3 is an increasing separation.

Theorem 6 *A permutation Π is stack sortable and pushpopable, if and only if Π has an increasing separation. Furthermore, if Π is a stack sortable and pushpopable permutation, then Π is an involution permutation.*

Proof. Suppose permutation Π is 231-avoid and 312-avoid. Cut Π into proper separation $\Pi_1, \Pi_2, \cdots, \Pi_k$ such that each Π_i is a decreasing subpermutation for any $1 \leq i \leq k$. For any $1 \leq i \leq k-1$, if there exist p_a in Π_i and p_b in Π_{i+1} such that $p_a > p_b$, then by the definition of Π_i and Π_{i+1}, there exists an integer c between a and b such that $p_b < p_a < p_c$ and p_c in Π_{i+1} (or $p_c < p_b < p_a$ and p_c in Π_i, resp.) and there is a (scattered) pattern 2-3-1 (or 3-1-2, respectively) in Π, a contradiction. Hence, Π has an increasing separation. It is clear that if a permutation has an increasing separation, then it must be an involution permutation. Since a stack sortable and pushpopable permutation is 231-avoid and 312-avoid, by the above reasoning it is an involution permutation.

In the other hand, if a permutation Π has an increasing separation $\Pi_1, \Pi_2, \cdots, \Pi_k$. Consider any three integers $a < b < c$ ($b < c < d$, respectively), if $p_b < p_c$, then p_b, p_c must in different decreasing subpermutations Π_i and Π_j with $i < j$. Hence, p_a and p_c (p_b and p_d, respectively) must in different decreasing subpermutations and $p_a < p_c$ ($p_b < p_d$, respectively) holds. That is, Π is 312-avoid (231-avoid, respectively) and Π is stack sortable and pushpopable by Lemmas 1 and 2. Q. E. D.

According to the definitions, given an increasing separation of $\Pi = \Pi_1 \circ \Pi_2 \circ \cdots \circ \Pi_k$ with the size of $\Pi_i = l_i \geq 1$ for $1 \leq i \leq k$, the inversion table of Π must be $01 \cdots (l_1-1) 01 \cdots (l_2-1) \cdots 01 \cdots (l_k-1)$. For example, the inversion table of $\Pi = 132654$ is 001012 since $k = 3$, $l_1 = 1$, $l_2 = 2$ and $l_3 = 3$. According to Theorem 6, a positive integer ordered k-tuple $(l_1, l_2, \cdots, l_k)$ is corresponding to an increasing separation of a permutation that is both stack sortable and pushpopable. That is, the number of permutations on $\{1, 2, \cdots, n\}$ which are both stack sortable and pushpopable is equal to the number of positive integer ordered k-tuples $(l_1, l_2, \cdots, l_k)$ where $l_1 + l_2 + \cdots + l_k = n$ for any possible $k \in \{1, 2, \cdots, n\}$. Because the answer of the latter is equal to

$$\sum_{k=1}^{n} C(n-k+k-1, k-1) = \sum_{k=0}^{n-1} C(n-1, k) = 2^{n-1}$$

where $C(n, k)$ is the binomial coefficient [3]. Therefore, the following theorem is obtained.

Theorem 7 *The number of permutations on $\{1, 2, \cdots, n\}$ which are both stack sortable and pushpopable is 2^{n-1}.*

Theorem 8 *$U_n = n! - 2C_n + 2^{n-1}$, where C_n is the Catalan number and U_n is the number of stack unsortable and unpushpopable permutations.*

Proof. According to stack sortable and pushpopable, it is clear that all permutations of length n can be classified into four types of permutations:

1. stack sortable and pushpopable permutations,
2. stack sortable and unpushpopable permutations,
3. stack unsortable and pushpopable permutations, and
4. stack unsortable and unpushpopable permutations.

It is already known that the number of stack sortable permutations of length n is C_n. Lemma 3 leads to the number of pushpopable permutations of length n is equal to C_n, too. By Theorem 7, the number of stack sortable and pushpopable permutations is 2^{n-1}. Thus, the number of stack sortable or pushpopable permutations is $2C_n - 2^{n-1}$. Consequently, the number of stack unsortable and unpushpopable permutations, namely U_n, is equal to $n! - 2C_n + 2^{n-1}$. Q. E. D.

Lemma 9 *There are no involution permutation is stack sortable but unpushpopable, or pushpopable but stack unsortable.*

Proof. For any involution permutation Π ($= \Pi^{-1}$), if Π is a stack sortable permutation, then Π^{-1} ($= (\Pi^{-1})^{-1} = \Pi$) is a pushpopable permutation by Lemma 3. In the same way, if Π is a pushpopable, then Π is also stack sortable. Q. E. D.

Denote the set of all stack sortable permutations by SS_n, pushpopable permutations by PP_n and involution permutations by $I(n)$. According to Lemma 9, the following two corollaries can be obtained directly.

Corollary 10 *$SS_n \cap PP_n = SS_n \cap I(n) = PP_n \cap I(n)$ for any positive integer n.*

Corollary 11 *$V_n = n! - 2C_n - I_n + 2^n$, where I_n is the number of involution permutations of length n and V_n is the number of stack unsortable, unpushpopable, and noninvolution permutations of length n.*

Table 1 shows the values of $n!$, 2^n, I_n, C_n, U_n, and V_n for $n = 1, 2, \cdots, 7$.

Table 1 Some Values of Number Sequences

n	1	2	3	4	5	6	7
$n!$	1	2	6	24	120	720	5040
2^n	2	4	8	16	32	64	128
I_n	1	2	4	10	26	76	232
C_n	1	2	5	14	42	132	429
U_n	0	0	0	4	52	488	4246
V_n	0	0	0	2	42	444	4080

Figure 2 shows the relationship among the sets of all permutations ($P(n)$), stack sortable permutations (SS_n), pushpopable permutations (PP_n), involution permutations ($I(n)$), and stack unsortable, unpushpopable, and noninvolution permutations ($V(n)$) of length n. Note that the set of stack unsortable and unpushpopable permutations is equal to $V(n) \cup I(n) - (\mathrm{SS}_n \cap \mathrm{PP}_n)$) and $U_n = V_n + I_n - 2^{n-1}$. And the exact number of each region in Figure 2 can be calculated.

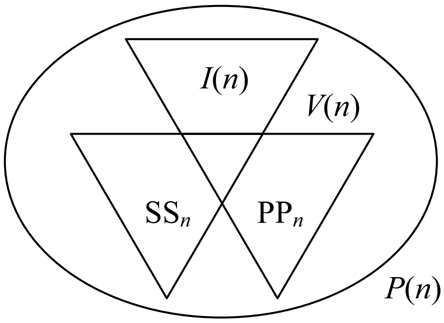

Figure 2: The relationship among $P(n)$, SS_n, PP_n, $I(n)$ and $V(n)$.

At last, to extend Algorithm A as a more powerful parallel algorithm that determines (or recognizes) a permutation to be one of the types SS_n, PP_n, $I(n)$, $V(n)$ or $\mathrm{SS}_n \cap \mathrm{PP}_n$ is not difficult, we skip the detail here.

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